Global Human Rights Disparities Analysis

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Relevant libaries import:

In [2]:
```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import random
import sympy as sp
from pivottablejs import pivot_ui
import hvplot.pandas
from bokeh.io import output_notebook
import seaborn as sns
```

Insight Question 1:

Between high-income countries and low/middle income countries, do people have different overall right to safety from state from 2017-2023? Specifically, do people's Physical Integrity Rights postively correlated with migrants/immigrants' risk of torture and ill-treatment?

Part 1: People in High-Income Countries have more Physical Integrity Rights

1. Import the dataset:

`hiy` is Income-Adjusted Data Sets of High Income Countries

`identity` is HRMI People at Risk (PaR) Dataset

`cpr` is HRMI Civil and Political Rights (CPR) Dataset

In [15]:
```
hiy = pd.read_csv('csv files/esr_hiy_incomeadjusted.csv')
identity = pd.read_csv('csv files/people_at_risk.csv')
cpr = pd.read_csv('csv files/cpr.csv')
```

2. Data processing

In the dataset `hiy`, the column High_Income_Country uses numeric labels:

- 1 represents High-Income Countries (HIY).
- 0 represents Low- and Middle-Income Countries (LMY).

3. Label the data

A new dataset `hiy_label` is created with 'Country' and 'High_Income_Country'.

In [16]:
```
hiy['High_Income_Country'] = hiy['High_Income_Country'].replace({1: 'HIY', 0: 'LMY'})
hiy = hiy.drop_duplicates(subset=['Country'])
hiy_label = hiy[['Country','High_Income_Country']]
display(hiy_label)
```

	Country	High_Income_Country
0	Aruba	HIY
32	Afghanistan	LMY
64	Angola	LMY
96	Albania	LMY
128	Andorra	HIY
...	...	...
6944	Kosovo	LMY
6976	Yemen, Rep.	LMY
7008	South Africa	LMY
7040	Zambia	LMY
7072	Zimbabwe	LMY

222 rows × 2 columns

4. Merge the data:

Merge the income classification labels `hiy_label` with the Civil and Political Rights dataset `cpr`, adding the income group (HIY / LMY) for each country. Extract variables related to Physical Integrity Rights (Physint).

- **Physint**: Physical Integrity Rights, which measure people's overall right to safety from the state.

In [17]:
```
cpr = cpr.copy()
hiy_label = hiy_label.copy()
cpr.rename(columns={'country': 'Country'}, inplace=True)
hiy_label.rename(columns={'High_Income_Country': 'Label'}, inplace=True)
label_merge = pd.merge(cpr, hiy_label, on='Country', how='inner')
display(label_merge)
cpr_phy = label_merge[['Country', 'year', 'physint_mean', 'physint_sd', 'physint_lo', 'physint_hi', 'Label
display(cpr_phy)
```

	Country	year	countryyear	iso3c	iso3n	cowcode	hrmicode	disap_mean	disap_sd	disap_lo	...	rel_hi
0	Angola	2017	AGO2017	AGO	24.0	540.0	1	6.464161	0.898106	5.314585	...	NaN
1	Angola	2018	AGO2018	AGO	24.0	540.0	1	7.174069	0.491192	6.545342	...	NaN
2	Angola	2019	AGO2019	AGO	24.0	540.0	1	6.357960	0.626105	5.556545	...	NaN
3	Angola	2020	AGO2020	AGO	24.0	540.0	1	5.453531	0.792786	4.438765	...	NaN
4	Angola	2021	AGO2021	AGO	24.0	540.0	1	4.963902	0.588297	4.210882	...	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...
177	Thailand	2023	THA2023	THA	764.0	800.0	47	8.162083	0.621875	7.366082	...	9.428472
178	Singapore	2022	SGP2022	SGP	702.0	830.0	49	8.410766	0.316615	8.005499	...	7.127632
179	Singapore	2023	SGP2023	SGP	702.0	830.0	49	8.403709	0.313198	8.002816	...	7.073044
180	Sri Lanka	2022	LKA2022	LKA	144.0	780.0	50	7.047200	0.292876	6.672319	...	6.533591
181	Sri Lanka	2023	LKA2023	LKA	144.0	780.0	50	7.356123	0.282373	6.994685	...	6.520242

182 rows × 52 columns

	Country	year	physint_mean	physint_sd	physint_lo	physint_hi	Label
0	Angola	2017	4.401868	0.795705	3.383366	5.420371	LMY
1	Angola	2018	5.122885	0.812793	4.082510	6.163259	LMY
2	Angola	2019	4.486554	0.790036	3.475308	5.497800	LMY
3	Angola	2020	3.438196	0.924271	2.255129	4.621263	LMY
4	Angola	2021	3.121782	0.849364	2.034596	4.208967	LMY
...	...	...	...	...	...	...	...
177	Thailand	2023	6.726162	0.901969	5.571642	7.880681	LMY
178	Singapore	2022	7.231867	0.765176	6.252442	8.211292	HIY
179	Singapore	2023	7.294268	0.743383	6.342738	8.245797	HIY
180	Sri Lanka	2022	5.090563	0.740136	4.143188	6.037937	LMY
181	Sri Lanka	2023	5.441030	0.729936	4.506712	6.375348	LMY

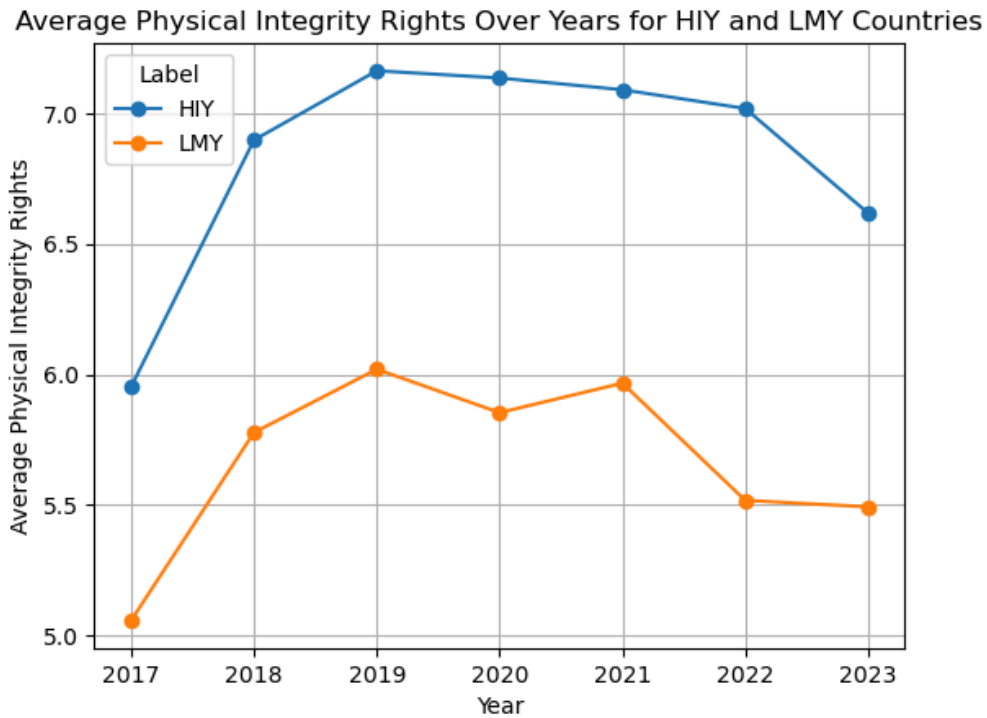
182 rows × 7 columns

5. Visualize the data

Create a line graph that shows the average Physical Integrity Rights (`physint_mean`) for high-income (HIY) and low/middle-income (LMY) countries over the years 2017–2023.

- X-axis: `year` (2017 to 2023)
- Y-axis: Average Physical Integrity Rights (`physint_mean`)
- Red Line: Represents HIY (High-Income Countries)
- Blue Line: Represents LMY (Low/Middle-Income Countries)

In [18]:
```
averaged_data = cpr_phy.groupby(['Label', 'year'])['physint_mean'].mean().reset_index()
hiy_data = averaged_data[averaged_data['Label'] == 'HIY']
lmy_data = averaged_data[averaged_data['Label'] == 'LMY']
plt.plot(hiy_data['year'], hiy_data['physint_mean'], label='HIY', marker='o')
plt.plot(lmy_data['year'], lmy_data['physint_mean'], label='LMY', marker='o')
plt.xlabel('Year')
plt.ylabel('Average Physical Integrity Rights')
plt.title('Average Physical Integrity Rights Over Years for HIY and LMY Countries')
plt.legend(title='Label')
plt.grid(True)
plt.show()
```



6. Observation

From the line graph, high-income countries (HIY) shows higher averaged physical integrity right than low/middle-income (LMY) countries across all different years. Thus, people in high-income countries overall have more right to safety from the state.

Part 2: In both HIY and LMY, migrant/immigrant risk of torture and ill-treatment show great variations.

1. Merge Data

Merge the HRMI People at Risk dataset `identity` with the income classification labels `hiy_label`, adding a classification of High-Income (HIY) or Low/Middle-Income (LMY) for each country.

2. Extract Migrant/Immigrant related Data

Extract variables related to migrants and immigrants `identity_imm`.

- `tort_atrisk_prop24`: Proportion of respondents who identified migrants/immigrants at risk of torture and ill-treatment.
- `tort_atrisk_count24`: Count of respondents identifying migrants/immigrants at risk of torture and ill-treatment.

In [19]:
```
hiy_label = hiy_label.copy()
hiy_label.rename(columns={'High_Income_Country': 'Label'}, inplace=True)
identity.rename(columns={'country': 'Country'}, inplace=True)
label_identity = pd.merge(identity, hiy_label, on='Country', how='inner')
display(label_identity)
identity_imm = label_identity[['Country', 'year', 'tort_atrisk_prop24', 'tort_atrisk_count24', 'Label']]
display(identity_imm)
```

	Country	year	countryyear	iso3c	iso3n	cowcode	hrmicode	food_total_atrisk_resp	food_atrisk_count1	food
0	Angola	2018	AGO2018	AGO	24.0	540.0	1	11	3	
1	Angola	2019	AGO2019	AGO	24.0	540.0	1	14	3	
2	Angola	2020	AGO2020	AGO	24.0	540.0	1	13	3	
3	Angola	2021	AGO2021	AGO	24.0	540.0	1	12	6	
4	Angola	2022	AGO2022	AGO	24.0	540.0	1	10	5	
...	...	...	...	...	...	...	...	...	...	
139	Maldives	2023	MDV2023	MDV	462.0	781.0	46	7	0	
140	Thailand	2022	THA2022	THA	764.0	800.0	47	13	4	
141	Thailand	2023	THA2023	THA	764.0	800.0	47	6	0	
142	Singapore	2023	SGP2023	SGP	702.0	830.0	49	7	1	
143	Sri Lanka	2023	LKA2023	LKA	144.0	780.0	50	29	12	

144 rows × 1359 columns

	Country	year	tort_atrisk_prop24	tort_atrisk_count24	Label
0	Angola	2018	0.615385	8.0	LMY
1	Angola	2019	0.562500	9.0	LMY
2	Angola	2020	0.071429	1.0	LMY
3	Angola	2021	0.333333	4.0	LMY
4	Angola	2022	0.090909	1.0	LMY
...	...	...	...	...	...
139	Maldives	2023	0.857143	6.0	LMY
140	Thailand	2022	0.071429	1.0	LMY
141	Thailand	2023	0.333333	2.0	LMY
142	Singapore	2023	0.142857	1.0	HIY
143	Sri Lanka	2023	0.068966	2.0	LMY

144 rows × 5 columns

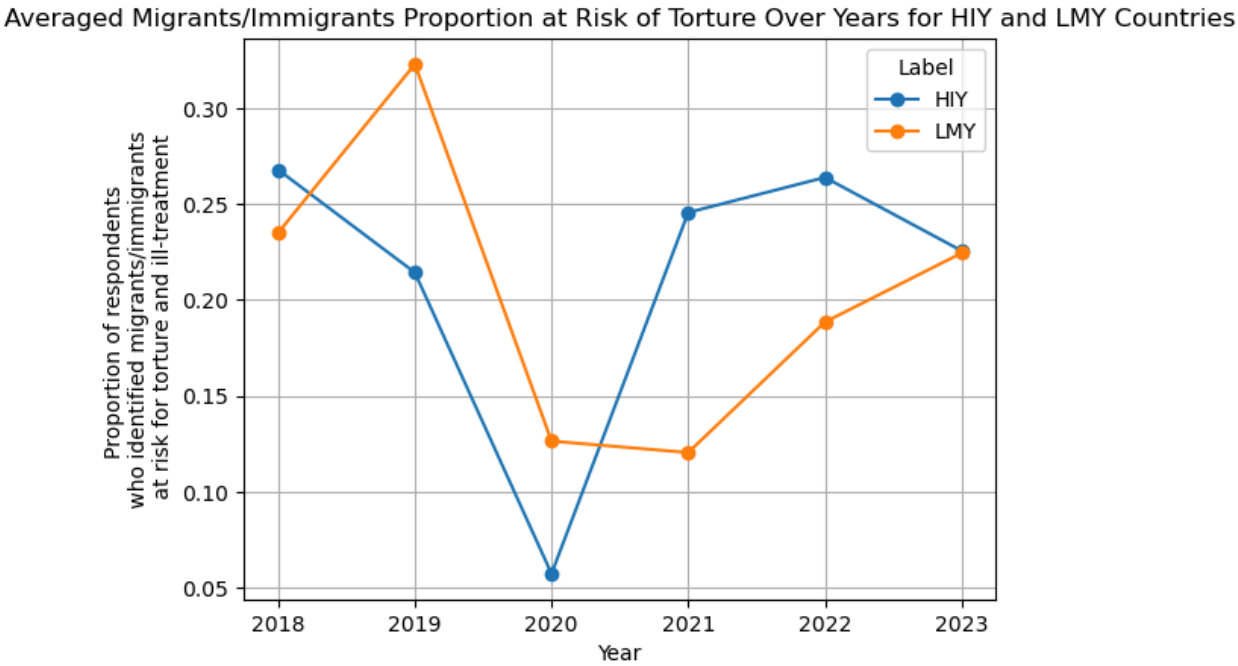
3. Visualize the data

Create a line graph to show the proportion of respondents who identified migrants/immigrants as being at risk of torture and ill-treatment in High-Income (HIY) and Low/Middle-Income (LMY) countries from 2017 to 2023.

- X-axis: `year` (2017 to 2023)
- Y-axis: The proportion of respondents who identified migrants/immigrants at risk for torture and ill-treatment (`tort_atrisk_prop24`).
- Red Line: Represents HIY (High-Income Countries)
- Blue Line: Represents LMY (Low/Middle-Income Countries)

In [20]:
```
averaged_imm = identity_imm.groupby(['Label', 'year'])['tort_atrisk_prop24'].mean().reset_index()
hiy_imm = averaged_imm[averaged_imm['Label'] == 'HIY']
lmy_imm = averaged_imm[averaged_imm['Label'] == 'LMY']
plt.plot(hiy_imm['year'], hiy_imm['tort_atrisk_prop24'], label='HIY', marker='o')
plt.plot(lmy_imm['year'], lmy_imm['tort_atrisk_prop24'], label='LMY', marker='o')
```

```
plt.xlabel('Year')
plt.ylabel('Proportion of respondents \n who identified migrants/immigrants \n at risk for torture and ill
plt.title('Averaged Migrants/Immigrants Proportion at Risk of Torture Over Years for HIY and LMY Countries
plt.legend(title='Label')
plt.grid(True)
plt.show()
```



4. Calculate standard deviation

Calculate the standard deviation (std) of the proportion of respondents `tort_atrisk_prop24` identifying migrants/immigrants as at risk of torture and ill-treatment for each income group (Label - HIY and LMY)to measure how much the values vary or spread out from the mean within each income group.

In [21]:
```
print(identity_imm.groupby('Label')['tort_atrisk_prop24'].std())
```

```
Label
HIY    0.215330
LMY    0.219511
Name: tort_atrisk_prop24, dtype: float64
```

5. Observation

The line plot shows that, between high-income (HIY) and low/middle-income (LMY) countries, there is no significant difference in the average proportion of migrants/immigrants at risk of torture and ill-treatment. This insignificant result may largely due to high variability and the influence of outliers within both groups:

- Both HIY and LMY groups show a standard deviation of approximately 0.2, which is substantial given the range of values (0.05 to 0.3). This large variability in responses within each group makes it difficult to detect meaningful differences between the two income groups.
- High standard deviation suggests the existence of outliers in both groups.

Outliers could disproportionately influence the overall averages.

Part 3: People in HIY have more physical integrity rights, and migrants/immigrants experience lower risks of torture and ill-treatment

1. Merge the data

Merge `cpr_phy` (Physical Integrity Rights data) with `identity_imm` (migrants/immigrants' risks). The new dataset after combination is called `final_merge`.

In [22]:
```
final_merge = pd.merge(cpr_phy, identity_imm, on='Country', how='inner')
display(final_merge)
```

	Country	year_x	physint_mean	physint_sd	physint_lo	physint_hi	Label_x	year_y	tort_atrisk_prop24	tort_atri
0	Angola	2017	4.401868	0.795705	3.383366	5.420371	LMY	2018	0.615385	
1	Angola	2017	4.401868	0.795705	3.383366	5.420371	LMY	2019	0.562500	
2	Angola	2017	4.401868	0.795705	3.383366	5.420371	LMY	2020	0.071429	
3	Angola	2017	4.401868	0.795705	3.383366	5.420371	LMY	2021	0.333333	
4	Angola	2017	4.401868	0.795705	3.383366	5.420371	LMY	2022	0.090909	
...	...	...	...	...	...	...	...	...	...	
815	Thailand	2023	6.726162	0.901969	5.571642	7.880681	LMY	2023	0.333333	
816	Singapore	2022	7.231867	0.765176	6.252442	8.211292	HIY	2023	0.142857	
817	Singapore	2023	7.294268	0.743383	6.342738	8.245797	HIY	2023	0.142857	
818	Sri Lanka	2022	5.090563	0.740136	4.143188	6.037937	LMY	2023	0.068966	
819	Sri Lanka	2023	5.441030	0.729936	4.506712	6.375348	LMY	2023	0.068966	

820 rows × 11 columns

2. Visualize the data

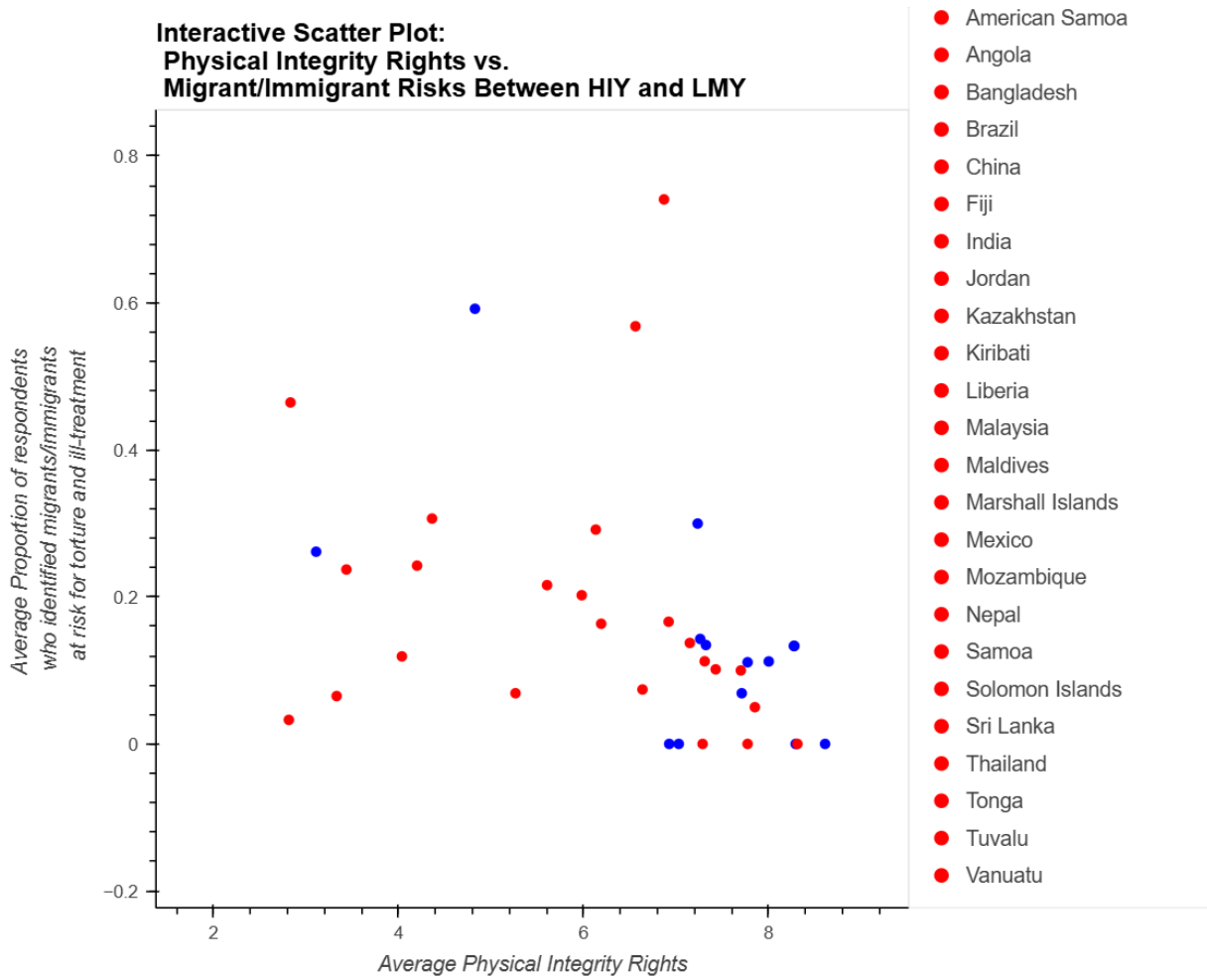
Create an interactive scatter plot showing the relationship between: `physint_mean` (Physical Integrity Rights) and `tort_atrisk_prop24` (Proportion of migrants/immigrants at risk of torture and ill-treatment), comparing high-income (HIY) and low/middle-income (LMY) countries.

- X-axis: Average Physical Integrity Rights (`physint_mean`).
- Y-axis: Proportion of migrants/immigrants at risk of torture (`tort_atrisk_prop24`).
- Blue dots represent HIY countries
- Red dots represent LMY countries.

In [23]:
```
scatter_data = final_merge.groupby(['Country', 'Label_x']).agg({
    'physint_mean': 'mean',
    'tort_atrisk_prop24': 'mean'
}).reset_index()

hiy_data = scatter_data[scatter_data['Label_x'] == 'HIY']
lmy_data = scatter_data[scatter_data['Label_x'] == 'LMY']

hiy_scatter = hiy_data.hvplot.scatter(
    x='physint_mean',
    y='tort_atrisk_prop24',
    by='Country',
    marker='o',
    color='blue',
    label='HIY',
    xlabel='Average Physical Integrity Rights',
    ylabel='Average Proportion of respondents \n who identified migrants/immigrants \n at risk for torture
)

lmy_scatter = lmy_data.hvplot.scatter(
    x='physint_mean',
    y='tort_atrisk_prop24',
```

```
    by='Country',
    marker='o',
    color='red',
    label='LMY',
    xlabel='Average Physical Integrity Rights',
    ylabel='Average Proportion of respondents \n who identified migrants/immigrants \n at risk for torture
)

final_plot = (hiy_scatter * lmy_scatter).opts(
    title='Interactive Scatter Plot: \n Physical Integrity Rights vs. \n Migrant/Immigrant Risks Between H
    legend_position='right',
    width=800,
    height=600
)

final_plot
```

Out[23]:



3. Observation

In the line graph comparing high-income (HIY) and low/middle-income (LMY) countries, there appears to be no significant distinction in the averaged proportion of migrants/immigrants at risk of torture and ill-treatment. This lack of clear distinction could be attributed to the significant variance within each income group. By averaging the countries in just two broad groups (HIY and LMY), we may overlook the heterogeneity and outliers within these groups.

In contrast, the scatter plot above zooms in the data for individual countries within each group, allowing us to reduce the influence of the large variation in the averages and provide a more precise evaluation. By plotting each country individually, between high-income and low/middle-income countries, there is a clear differences in the pattern of average physical integrity rights and propotion of immigrants/migrants at risk of torture and ill-treatment.

For **high income countries**: HIY countries cluster mostly in the lower-right corner of the scatter plot. In HIY countries, people have more physical integrity rights (`physint_mean`), which corresponds to a lower proportion of migrants/immigrants at risks of torture and ill-treatment (`tort_atrisk_prop24`). Thus, in HIY countries, there is a negative correlation between people's physical integrity rights and immigrants/migrants' risks of torture and ill-treatment.

For **low income countries**: LMY countries show great variability, with points scattered across the graph. This reflects wide disparities in physical integrity rights and immigrants/migrants' risks of torture and ill-treatment. Thus, in low/middle-income countries, there is no correlation between people's physical integrity rights and immigrants/migrants' risks of torture and ill-treatment.

However, two **outliers** are identified in HIY:

- The United States:

Located at approximately (4.8, 0.6), it exhibits a lower-than-expected physical integrity score and higher risks for migrants/immigrants compared to other HIY countries.

- Saudi Arabia:

Found at (3.1, 0.3), it has a low physical integrity score and a moderate risk proportion, deviating from the general HIY trend.

Insight 1 :

From 2017 to 2023, people in high-income countries (HIY) overall have more rights to safety from state compared to people in low/middle income countries. A stronger protection for physical integrity rights in high-income countries also correlates to lower proportions of migrants/immigrants at risks of torture and ill-treatment, indicating a negative correlation. Thus, in high-income countries, stronger overall protections for physical integrity is related with better safety and rights for migrants/immigrants. In contrast, in low/middle-income countries, there is a considerable variability in both physical integrity rights and the risks faced by migrants/immigrants. Thus, individuals in low/middle-income countries are lack of consistent protection, and migrants are exposed to varying degrees of risk for torture and ill-treatment. This variability reflects governance challenges and the unequal provision of physical integrity rights, leading to inconsistent safety for migrants/immigrants in low/middle-income countries.

Insight Question 2:

From 2017 to 2023, which human right is the most often denied or poorly protected in pacific countries vs. continental countries?

1. Import the dataset

`pacific_taiwan_data_df` is Data Sources used for Taiwan and Pacific Countries
`cpr_df` is HRMI Civil and Political Rights (CPR) Dataset

In [24]:
```
cpr_df = pd.read_csv('csv files/cpr.csv')
pacific_taiwan_data_df = pd.read_csv('csv files/Pacific_Taiwan_data.csv', encoding='latin1')
```

2. Label the data

Add a binary column 'pacific_binary' to `cpr_df`:

- 1 indicates the country is in the Pacific region.
- 0 indicates the country is not in the Pacific region.

In [25]:
```
pacific_countries = pacific_taiwan_data_df['Country'].str.strip().unique()
cpr_df['pacific_binary'] = cpr_df['country'].apply(lambda x: 1 if x in pacific_countries else 0)
display(cpr_df)
```

	country	year	countryyear	iso3c	iso3n	cowcode	hrmicode	disap_mean	disap_sd	disap_lo	...	rel_hi
0	Angola	2017	AGO2017	AGO	24.0	540.0	1	6.464161	0.898106	5.314585	...	NaN
1	Angola	2018	AGO2018	AGO	24.0	540.0	1	7.174069	0.491192	6.545342	...	NaN
2	Angola	2019	AGO2019	AGO	24.0	540.0	1	6.357960	0.626105	5.556545	...	NaN
3	Angola	2020	AGO2020	AGO	24.0	540.0	1	5.453531	0.792786	4.438765	...	NaN
4	Angola	2021	AGO2021	AGO	24.0	540.0	1	4.963902	0.588297	4.210882	...	NaN
...	...	...	...	...	...	...	...	...	...	...	...	...
225	Thailand	2023	THA2023	THA	764.0	800.0	47	8.162083	0.621875	7.366082	...	9.428472
226	Singapore	2022	SGP2022	SGP	702.0	830.0	49	8.410766	0.316615	8.005499	...	7.127632
227	Singapore	2023	SGP2023	SGP	702.0	830.0	49	8.403709	0.313198	8.002816	...	7.073044
228	Sri Lanka	2022	LKA2022	LKA	144.0	780.0	50	7.047200	0.292876	6.672319	...	6.533591
229	Sri Lanka	2023	LKA2023	LKA	144.0	780.0	50	7.356123	0.282373	6.994685	...	6.520242

230 rows × 52 columns

3. Select the data

Dynamically select all columns that include 'mean' in their name, which represent the average scores for different human rights.

- `Disap` : Right to freedom from disappearance
- `Exkill` : Right to freedom from extrajudicial execution
- `Arrest` : Right to freedom from arbitrary or political arrest and imprisonment
- `Tort` : Right to freedom from torture and ill-treatment
- `Depex` : Right to freedom from death penalty execution
- `Express` : Right to opinion and expression
- `Polpart` : Right to participate in government
- `Assem` : Right to assembly and association
- `Rel` : Right to freedom of religion and belief
- `Physint` : Overall right to safety from the state, i.e. Physical Integrity Rights
- `Empower` : Overall right to empowerment

In [26]:
```
rights_columns = [col for col in cpr_df.columns if 'mean' in col]
display(rights_columns)
```

```
['disap_mean',
 'exkill_mean',
 'arrest_mean',
 'tort_mean',
 'dpex_mean',
 'express_mean',
 'polpart_mean',
 'assem_mean',
 'rel_mean',
 'physint_mean',
 'empower_mean']
```

4. Calculation

Calculate the average values of the selected human rights columns, grouped by `pacific_binary` and `year`.

In [27]:
```
grouped_means = cpr_df.groupby(['pacific_binary', 'year'])[rights_columns].mean().reset_index()
display(grouped_means)
```

	pacific_binary	year	disap_mean	exkill_mean	arrest_mean	tort_mean	dpex_mean	express_mean	polpart_mean
0	0	2017	6.733320	6.247352	5.245753	4.890763	9.230942	4.675750	4.603258
1	0	2018	6.735316	6.199742	5.115282	4.893804	9.173853	4.625586	5.073367
2	0	2019	6.911426	6.292821	4.920348	5.026125	9.195390	4.578641	5.143065
3	0	2020	6.805790	6.411092	5.212349	4.964187	8.866773	4.335139	5.019546
4	0	2021	6.749326	6.534168	5.391714	5.196561	8.971095	4.411066	5.088646
5	0	2022	6.827311	6.611590	5.205224	5.132033	8.844059	4.422194	5.056482
6	0	2023	6.892017	6.650289	5.085363	5.054111	8.794397	4.508944	4.993601
7	1	2017	8.292837	7.052309	5.469275	4.838824	10.000000	2.841465	4.365599
8	1	2018	8.554965	8.021153	6.894804	6.795908	10.000000	5.856057	6.803963
9	1	2019	8.621577	8.031246	7.357238	7.071760	10.000000	6.240824	7.021942
10	1	2020	8.420540	8.169713	7.810239	7.158492	9.885901	6.118137	7.015026
11	1	2021	8.492205	8.502044	8.006396	7.448673	10.000000	6.510010	7.129258
12	1	2022	8.402668	8.419019	7.961962	6.805907	10.000000	6.253374	6.465792
13	1	2023	8.377878	8.366530	7.955499	5.973340	10.000000	7.621193	6.639057

5. Combine the data

Reshape the data into a long format that combine all human rights into one column `right`.

Identify the human right with the lowest mean value for each combination of `pacific_binary` and `year`.

In [28]:
```
lowest_means = grouped_means.melt(id_vars=['pacific_binary', 'year'], var_name='right', value_name='right_
lowest_means_per_year = lowest_means.loc[lowest_means.groupby(['pacific_binary', 'year'])['right_mean'].ic
lowest_means_per_year['right'] = lowest_means_per_year['right'].replace({
    'express_mean': 'express',
    'empower_mean': 'empower',
    'assem_mean': 'assem',
    'exkill_mean': 'exkill',
    'disap_mean': 'disap',
    'arrest_mean': 'arrest',
    'tort_mean': 'tort',
    'dpex_mean': 'dpex',
    'rel_mean': 'rel',
    'physint_mean': 'physint'
```

```
})
display(lowest_means)
display(lowest_means_per_year)
```

	pacific_binary	year	right	right_mean
0	0	2017	disap_mean	6.733320
1	0	2018	disap_mean	6.735316
2	0	2019	disap_mean	6.911426
3	0	2020	disap_mean	6.805790
4	0	2021	disap_mean	6.749326
...	...	...	...	...
149	1	2019	empower_mean	6.901580
150	1	2020	empower_mean	6.544884
151	1	2021	empower_mean	6.669821
152	1	2022	empower_mean	6.443977
153	1	2023	empower_mean	7.247096

154 rows × 4 columns

	pacific_binary	year	right	right_mean
140	0	2017	empower	4.579556
71	0	2018	express	4.625586
72	0	2019	express	4.578641
73	0	2020	express	4.335139
144	0	2021	empower	4.375018
103	0	2022	assem	4.396123
146	0	2023	empower	4.387060
77	1	2017	express	2.841465
78	1	2018	express	5.856057
79	1	2019	express	6.240824
80	1	2020	express	6.118137
81	1	2021	express	6.510010
82	1	2022	express	6.253374
55	1	2023	tort	5.973340

6. Split the data

Split the data for plotting into two subsets:

- `pacific_data` : Contains Pacific countries.
- `continental_data` : Contains non-Pacific (continental) countries.

In [29]:
```
pacific_data = lowest_means_per_year[lowest_means_per_year['pacific_binary'] == 1]
continental_data = lowest_means_per_year[lowest_means_per_year['pacific_binary'] == 0]
display(pacific_data)
display(continental_data)
```

	pacific_binary	year	right	right_mean
77	1	2017	express	2.841465
78	1	2018	express	5.856057
79	1	2019	express	6.240824
80	1	2020	express	6.118137
81	1	2021	express	6.510010
82	1	2022	express	6.253374
55	1	2023	tort	5.973340

	pacific_binary	year	right	right_mean
140	0	2017	empower	4.579556
71	0	2018	express	4.625586
72	0	2019	express	4.578641
73	0	2020	express	4.335139
144	0	2021	empower	4.375018
103	0	2022	assem	4.396123
146	0	2023	empower	4.387060

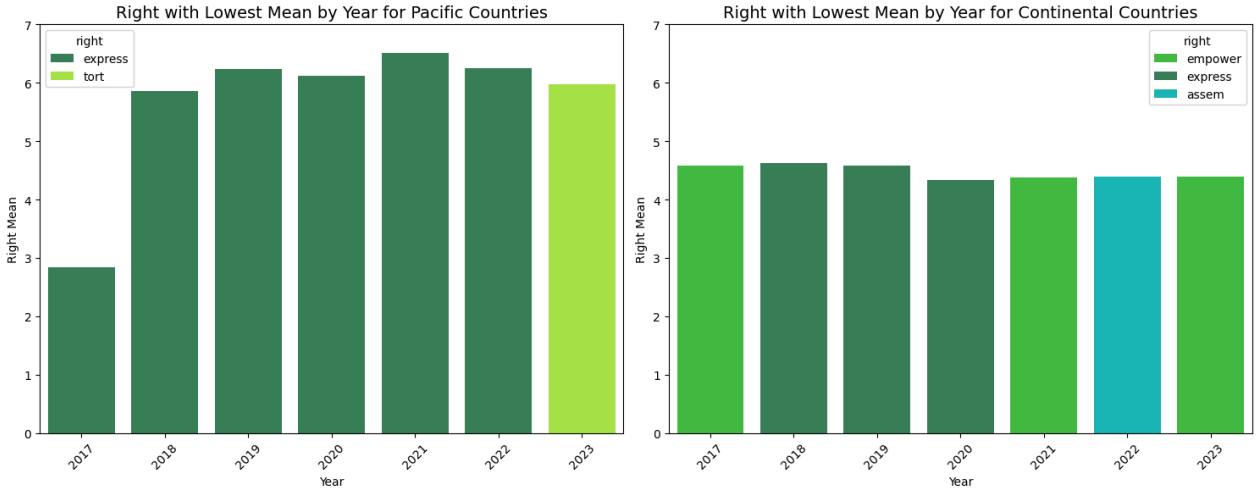
7. Visualize the data

Create two bar plost showing how the human rights with the lowest average scores evolved over this period for Pacific and Continental countries:

- X-axis: `year`
- Y-axis: `right_mean` (the lowest mean value per year for each group).

In [30]:
```
# Create a figure with two subplots (1 row, 2 columns)
fig, axes = plt.subplots(1, 2, figsize=(15, 6))

custom_palette = {
    'express': 'seagreen',
    'empower': 'limegreen',
    'assem': 'darkturquoise',
    'tort': 'greenyellow'
}

# Plot for HIY countries (pacific_binary = 1) in the first subplot
sns.barplot(x='year', y='right_mean', hue='right', data=pacific_data, ax=axes[0], palette=custom_palette)
axes[0].set_title('Right with Lowest Mean by Year for Pacific Countries', fontsize=14)
axes[0].set_xlabel('Year')
axes[0].set_ylabel('Right Mean')
axes[0].tick_params(axis='x', rotation=45)
axes[0].set_ylim(0, 7)

# Plot for LMY countries (pacific_binary = 0) in the second subplot
sns.barplot(x='year', y='right_mean', hue='right', data=continental_data, ax=axes[1], palette=custom_palet
axes[1].set_title('Right with Lowest Mean by Year for Continental Countries', fontsize=14)
axes[1].set_xlabel('Year')
axes[1].set_ylabel('Right Mean')
axes[1].tick_params(axis='x', rotation=45)
axes[1].set_ylim(0, 7)

# Adjust layout to avoid overlap
plt.tight_layout()
```

```
# Show the combined plot
plt.show()
```



8. Observation

The graph highlights notable disparities between Pacific and Continental countries in terms of their lowest-scoring human rights over time.

Pacific countries: the lowest mean score in 2017 was rights of expression freedom (`express`) with a value slightly above 3, which indicates significant challenges in rights of opinion and expression. However, there is a significant improvement in right of expression from 2017 to 2018, and the average of expression right remains relative high from then on (around 6). Then, by 2023, the lowest-scoring right shifted to freedom from torture and ill-treatment (`tort`), which also has a relatively high score (around 7). This progression reflects significant improvement made by Pacific countries in addressing their weakest human rights issues.

Continental countries: continental countries show no significant improvements in their lowest right score, and the lowest score stabilizes around 4.5 over years. There is also a fluctuation bewteen rights with the lowest score over years. In 2017, empowerment (`empower`) was the lowest-scoring right, with a mean score around 4. Between 2018 to 2020, freedom of expression emerged as the lowest-scoring right, with scores stabilizing around 5, indicates significant challenges in rights of opinion and expression. By 2022, assembly rights (`assem`) became the lowest-scoring right, indicating challenges in guaranteeing the right to assembly and association. In 2023, empowerment once again became the lowest-scoring right, showing persistent barriers to individual empowerment.

Pacific countries show more significant progresses overtime, with their lowest-scoring rights improving from ~3 in 2017 to nearly 7 in 2023. In contrast, continental countries showed slower improvements, with their lowest-scoring rights stabilizes around 4.5 throughout the years. Continental countries also have greater fluctuations between rights with the lowest score, suggesting more rights that are at risks/with low scores.

Insight 2 :

In **pacfic countries**, **rights of expression freedom** is the most poorly protected human right, appearing as the lowest-scoring right for 5 years (between 2017 and 2022). In 2023, **freedom from torture and ill-treatment** became the lowest-scoring right. In **continental countries**, both the **empowerment right** and the **right of expression freedom** are the most denied human righs, each appears as the lowest right for three years. **Assembly right** is also poorly protected and appear as the lowest-scoring right in 2022.

When comparing the two regions, pacific countries made more significant improvements in protecting human rights compared to the continental countries. Pacific countries also have less fluctuations in their lowest-scoreing human rights compared to continental countries, suggesting fewer rights at risks or with low values. Thus, individuals in pacific countries generally have better protected human rights compared to continental countries.

Insight Question 3:

What is the status of empowerment rights in each Asia-Pacific regions between 2018 to 2023? Specifically, does women in regions with stronger empowerment rights have greater rights in education?

1. Import the dataset

`people_at_risk_df` is People At Risk Dataset
`cpr_df` is HRMI Civil and Political Rights (CPR) Dataset

In [3]:
```
cpr_df = pd.read_csv('csv files/cpr.csv')
people_at_risk_df = pd.read_csv('csv files/people_at_risk.csv')
```

2. Filter the data

Filter the data to include only the selected Asia-Pacific regions countries (listed in `target_countries`). Also extract relevant columns including:

- Empower: Overall right to empowerment
- `educ_atrisk_prop10`: Proportion of women/girls at risk of lacking right to education

In [11]:
```
target_countries = ['China', 'South Korea', 'HongKong', 'Taiwan', 'Vietnam', 'Malaysia', 'Thailand', 'Sing
filtered_cpr_df = cpr_df[(cpr_df['country'].isin(target_countries))]
filtered_cpr_df = filtered_cpr_df[['country', 'year', 'empower_mean']]
filtered_people_at_risk_df = people_at_risk_df[(people_at_risk_df['country'].isin(target_countries))]
filtered_people_at_risk_df = filtered_people_at_risk_df[['country', 'year', 'educ_atrisk_prop10']]
display(filtered_cpr_df)
display(filtered_people_at_risk_df)
```

	country	year	empower_mean
104	South Korea	2017	7.091490
105	South Korea	2018	7.096694
106	South Korea	2019	6.599675
107	South Korea	2020	6.437470
108	South Korea	2021	7.256850
109	South Korea	2022	7.158103
110	South Korea	2023	5.785240
125	Vietnam	2017	1.207825
126	Vietnam	2018	2.108225
127	Vietnam	2019	2.430420
128	Vietnam	2020	2.496144
129	Vietnam	2021	2.357853
130	Vietnam	2022	2.212683
131	Vietnam	2023	2.257634
199	Malaysia	2019	5.672837
200	Malaysia	2020	3.773163
201	Malaysia	2021	4.088863
202	Malaysia	2022	4.383113
203	Malaysia	2023	4.775669
204	Taiwan	2019	7.072268
205	Taiwan	2020	6.988512
206	Taiwan	2021	6.993072
207	Taiwan	2022	7.069246
208	Taiwan	2023	7.247096
209	China	2020	1.613400
210	China	2021	1.516172
211	China	2022	1.627610
212	China	2023	1.913970
223	Thailand	2021	3.681815
224	Thailand	2022	4.080241
225	Thailand	2023	4.842543
226	Singapore	2022	4.089519
227	Singapore	2023	4.190406

	country	year	educ_atrisk_prop10
89	South Korea	2018	0.000000
90	South Korea	2019	0.000000
91	South Korea	2020	0.000000
92	South Korea	2021	0.000000
93	South Korea	2022	0.000000
94	South Korea	2023	0.125000
107	Vietnam	2018	0.235294
108	Vietnam	2019	0.153846
109	Vietnam	2020	0.000000
110	Vietnam	2021	0.200000
111	Vietnam	2022	0.052632
112	Vietnam	2023	0.000000
163	Malaysia	2020	0.200000
164	Malaysia	2021	0.166667
165	Malaysia	2022	0.333333
166	Malaysia	2023	0.166667
167	Taiwan	2020	0.000000
168	Taiwan	2021	0.033333
169	Taiwan	2022	0.040000
170	Taiwan	2023	0.066667
171	China	2021	0.243243
172	China	2022	0.145833
173	China	2023	0.090909
181	Thailand	2022	0.076923
182	Thailand	2023	0.333333
183	Singapore	2023	0.000000

3. Merge the data

Merges the filtered datasets `filtered_cpr_df` and `filtered_people_at_risk_df` on `country` and `year` to create a new dataset `merged_df`.

In [12]:
```
merged_df = pd.merge(filtered_cpr_df, filtered_people_at_risk_df, on=['country', 'year'])
display(merged_df)
```

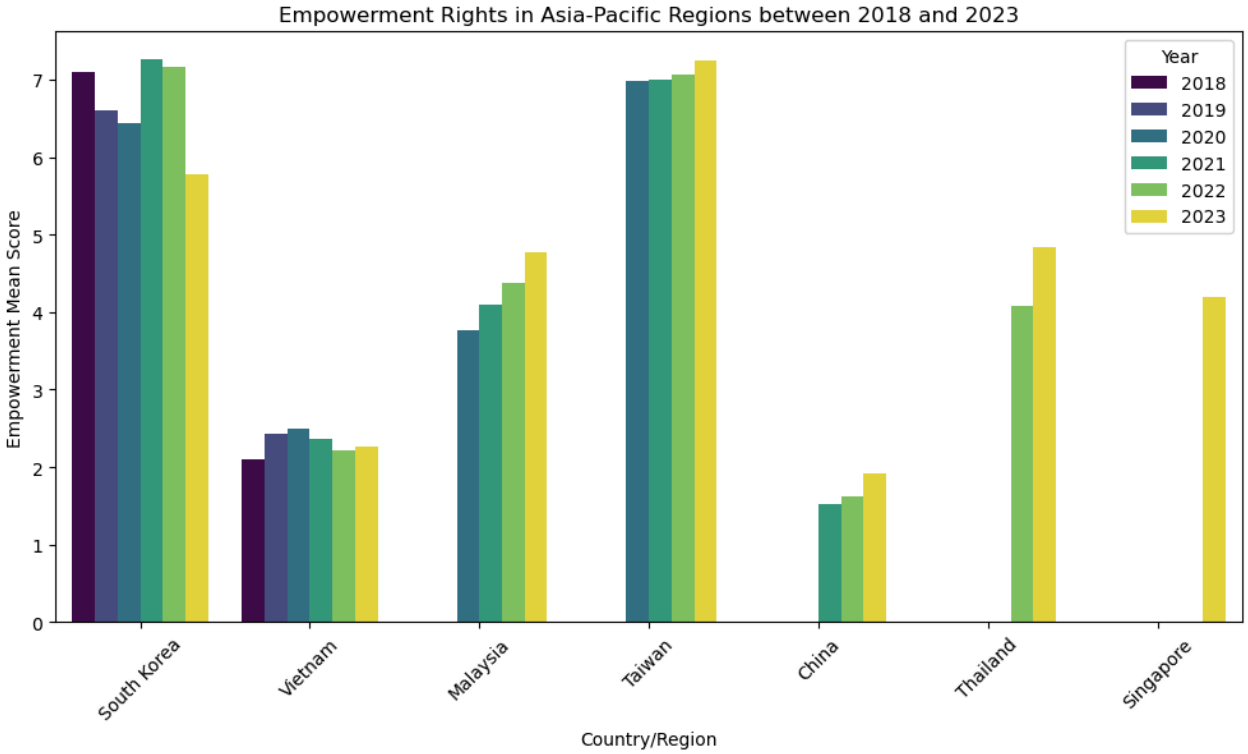
	country	year	empower_mean	educ_atrisk_prop10
0	South Korea	2018	7.096694	0.000000
1	South Korea	2019	6.599675	0.000000
2	South Korea	2020	6.437470	0.000000
3	South Korea	2021	7.256850	0.000000
4	South Korea	2022	7.158103	0.000000
5	South Korea	2023	5.785240	0.125000
6	Vietnam	2018	2.108225	0.235294
7	Vietnam	2019	2.430420	0.153846
8	Vietnam	2020	2.496144	0.000000
9	Vietnam	2021	2.357853	0.200000
10	Vietnam	2022	2.212683	0.052632
11	Vietnam	2023	2.257634	0.000000
12	Malaysia	2020	3.773163	0.200000
13	Malaysia	2021	4.088863	0.166667
14	Malaysia	2022	4.383113	0.333333
15	Malaysia	2023	4.775669	0.166667
16	Taiwan	2020	6.988512	0.000000
17	Taiwan	2021	6.993072	0.033333
18	Taiwan	2022	7.069246	0.040000
19	Taiwan	2023	7.247096	0.066667
20	China	2021	1.516172	0.243243
21	China	2022	1.627610	0.145833
22	China	2023	1.913970	0.090909
23	Thailand	2022	4.080241	0.076923
24	Thailand	2023	4.842543	0.333333
25	Singapore	2023	4.190406	0.000000

4. Visualize the data -- Bar chart of Empowerment Rights in Asia-Pacific Regions

Plots a bar chart comparing `empower_mean` scores for countries/regions across different years to visualize changes in empowerment rights over time for each country.

- X-axis: `country`
- Y-axis: Empowerment mean scores `empower_mean`.

In [13]:
```
plt.figure(figsize=(12, 6))
sns.barplot(data=merged_df, x='country', y='empower_mean', hue='year', palette='viridis')
plt.title('Empowerment Rights in Asia-Pacific Regions between 2018 and 2023')
plt.ylabel('Empowerment Mean Score')
plt.xlabel('Country/Region')
plt.xticks(rotation=45)
plt.legend(title='Year')
plt.show()
```



5. Observation

The bar chart compares the empowerment scores of Asia-Pacific regions between 2018 and 2023. Taiwan and South Korea rank the highest in empowerment scores among the regions, having average empowerment scores around 7 and 6.5 respectively. Malaysia follow, with an improving empowerment scores from 4 to 5 overtime. Vietnam and China exhibit significantly lower empowerment scores (around 2). Interestingly, while both Vietnam and China are communist countries, Taiwan and South Korean are capitalist regions. This difference in political and economic systems may be a potential factor impacting people's empowerment rights. The data is missing for Thailand and Singapore from 2018 to 2021, posing a difficulty in analyzing their long term trend. However, both countries tend to have similar scores as Malasysia (arounf 4.5), suggesting a similarity shared between capitalist countries in Southeast Asia. Due to the disparity in empowerment scores between capitalist countries/regions and communist countries, further explorations into the impact of governance systems on empowerment scores may provide meaningful insights.

6. Scatter Plot -- Empowerment vs. Education Rights at Risk in Asia-Pacific Regions

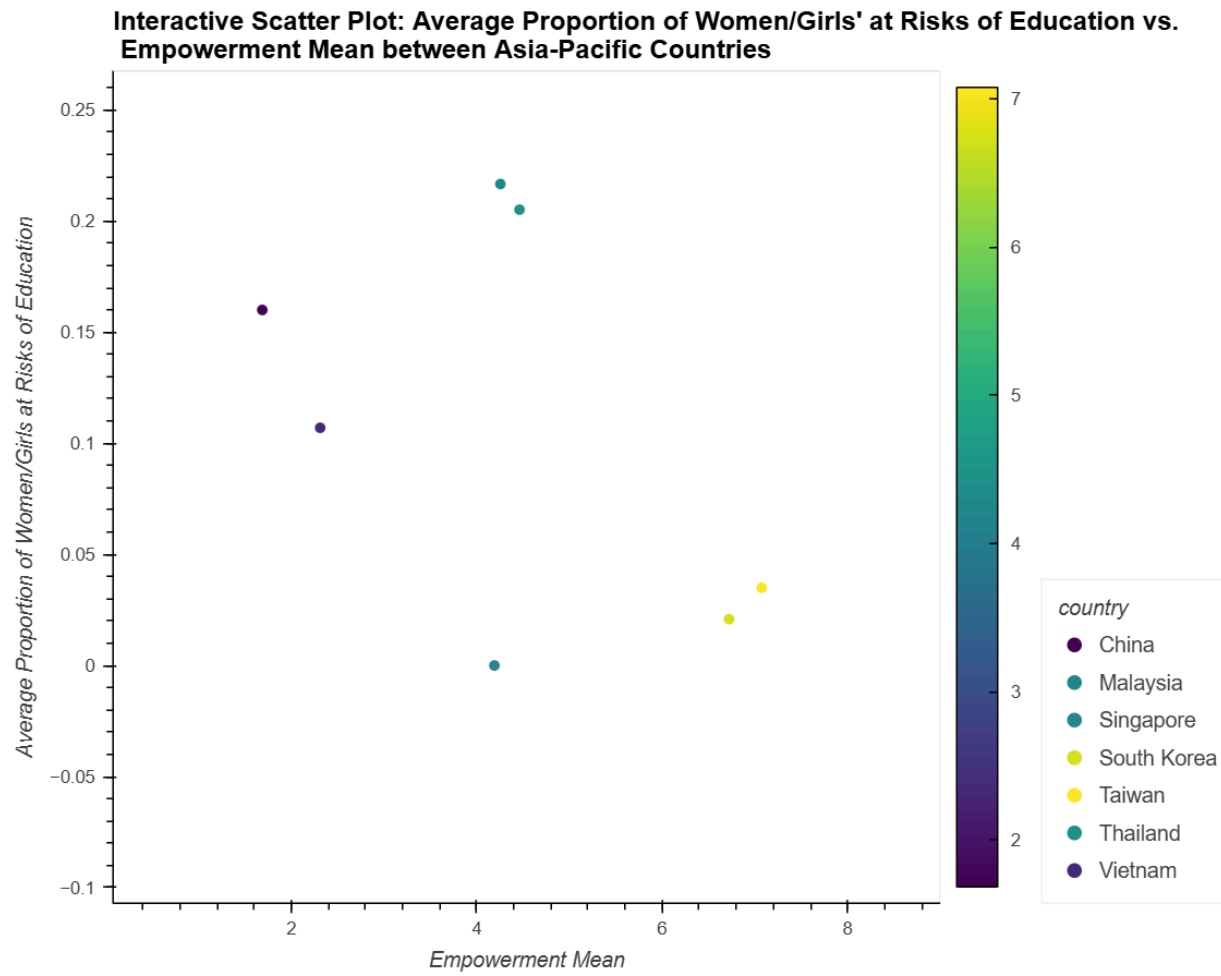
Plots a scatter plot showing the relationship between empowerment rights `empower_mean` and work rights at risk `educ_atrisk_prop10`. Points are distinguished by country and year to analyze the correlation between empowerment scores and perceived risks to work rights.

- X-axis: Empowerment mean scores `empower_mean`.
- Y-axis: Total respondents identifying work rights at risk `educ_atrisk_prop10`.

In [14]:
```
scatter_data = merged_df.groupby('country').agg({
    'empower_mean': 'mean',
    'educ_atrisk_prop10': 'mean'
}).reset_index()

# Create the scatter plot using the 'hvplot.scatter' function
scatter = scatter_data.hvplot.scatter(
```

```
    x='empower_mean',
    y='educ_atrisk_prop10',
    by='country',
    marker='o',
    c='empower_mean',  # Color points based on 'empower_mean'
    cmap='viridis',  # Apply 'viridis' colormap
    label='country',
    ylabel='Average Proportion of Women/Girls at Risks of Education',
    xlabel='Empowerment Mean'
)

# Customize the plot options
final_plot = scatter.opts(
    title="Interactive Scatter Plot: Average Proportion of Women/Girls' at Risks of Education vs. \n Empov
    legend_position='right',
    width=800,
    height=600
)

# Display the final plot
final_plot
```

Out[14]:



7. Observation

From the scatter plot, there is a **negative correlation** between empowerment mean and the proportion of women/girls at risks of education. A higher empowerment generally associated with fewer risks at education for women/girls. Countries/regions with higher empowerment scores (e.g., Taiwan, South Korea, and singapore), show lower proportions of women/girls at education risks, with values below 0.1.

This indicates that higher empowerment may be associated with stronger policies and societal structures that mitigate women/girls' risks in education. In contrast, China and Vietnam, with lower empowerment scores, exhibit higher proportions of women/girls at education risks (above 0.1). Interestingly, Malaysia and Singapore, having a moderate empowerment mean scores (~4-5), exhibit a high proportion of women/girls at education risks (~0.05-0.1). This discrepency from the general trend suggest the existence of other confounding factors influencing women/girls' risks at education (e.g., economic development, governance systems, cultural norms).

Insight 3 :

In Asia-pacific regions, capitalist countries/regions generally have higher empowerment rights than communist countries, therefore, political and economic systems may be a potential factor impacting people's empowerment rights. Moreover, a negative correlation is observed between empowerment score and women/girls' risks in education. A higher empowerment score is related to lower risks of women/girls in education. In countries with weaker empowerment rights, women or girls are more likely to experience systemic barriers such as gender inequality or lack of resources, exposing them to higher risks at education. Combining with previous observation that capitalist regions generally have higher empowerment rights, this observation may be shaped by socio-economic and political dynamics in different countries.